Australian Journal of Earth Sciences (2011) **58**, (361–374)

Magnetic characteristics of the Hiltaba Suite Granitoids and Volcanics: Late Devonian overprinting and related thermal history of the Gawler Craton

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The ~1590 Ma Gawler Range Volcanics (GRV) and 1595–1575 Ma Hiltaba Suite Granitoids (HSG), from the central Gawler Craton of South Australia, can be divided into an oxidised Roxby Downs-type containing hematite and magnetite, and a less oxidised Kokatha-type that commonly contains ilmenite and titanomagnetite. In both types the predominant magnetic phase is pure end-member magnetite, except for a minor amount of titanomagnetite present in the Kokatha-type Black Yardea Dacite. The characteristic remanence observed in the GRV is directed upward to the northeast. A fold test is strongly negative, indicating that this characteristic remanence is a post-folding overprint although the age of folding is uncertain. Rock magnetic properties identify magnetically unstable multidomain magnetite particles. The magnetisation has been blocked as the crust cooled from above multidomain (MD) unblocking temperatures probably during a mid-phase of the Alice Springs Orogeny (ASO). The *in situ* mean direction, not corrected for tilt, from 69 samples is declination = 22.6°, inclination = −44.9°, α_{95} = 6.9° and the corresponding pole position is latitude λ_P = 69.5°S, longitude φ_P = 36.1°E, dp = 5.5°, dm = 8.7°. No directions of opposite polarity were found and no consistent remanence was found in either type of the HSG. The pole position from GRV is consistent with other Australian paleopoles from rocks of about 360–370 Ma. Apatite fission track ages (AFTA) for the Gawler Craton have yielded a mean age of 331 ± 30 Ma. It is likely that the remanence and fission tracks have been reset during exhumation/cooling after the ASO. The younger AFTA of ~330 Ma are consistent with the track retention temperature (100 ± 20°C) being attained after the higher remanence blocking temperature of ~>200°C. The apatite ages are consistent with thermal modelling of $^{40}Ar/^{39}Ar$ data from K-feldspars from the nearby Mount Painter Province.

KEY WORDS: South Australia, Gawler Range Volcanics, Hiltaba Suite Granitoids, Alice Springs Orogeny, paleomagnetism, fission track dating

INTRODUCTION

The ~1590 Ma Gawler Range Volcanics (GRV) and the 1595–1575 Ma Hiltaba Suite Granitoids (HSG) outcrop in the central Gawler Craton of South Australia. They produce characteristic, although quite distinctive, magnetic anomalies and can be easily mapped beneath cover. The volcanics typically produce very active magnetic patterns, while the patterns above granitoids range from subdued to remarkably flat and smooth. The HSG and GRV can be subdivided into two types, each correlated with a distinctive mineral association. The more oxidised members are of the Roxby Downs-type and are associated with Fe-oxide-Cu–Au mineralisation, while the less oxidised Kokatha-types are associated with vein Au (±Sn±Ag) mineralisation (Budd *et al.* 1998). Here we report magnetic properties of the Hiltaba Suite Granitoids and the Gawler Range Volcanics.

Magnetic remanences and susceptibilities of samples were measured to characterise the different assemblages and assist mineral exploration and geological interpretation of magnetic surveys over the Gawler Craton. In addition, standard paleomagnetic methods were applied to determine the characteristic stable remanences present to place constraints on geological events that have affected the Gawler craton.

GEOLOGICAL SETTING

The evolution of the Gawler craton began in the late Archean (>2500 Ma) with formations represented by meta-sediments interlayered with banded iron formations, carbonate and siliceous rocks, and felsic and mafic–ultramafic volcanics including komatiites (Hand *et al.* 2007). The volcanics have been interpreted (Swain *et al.* 2005) to have formed in an active subduction environment. The dominant U–Pb detrital zircon ages for the metasediments range between *ca* 2850 and 2510 Ma with minor older zircons ranging between *ca* 3150

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ISSN 0812-0099 print/ISSN 1440-0952 online © 2011 Geological Society of Australia
DOI: 10.1080/08120099.2010.549239

and 2950 Ma (Swain *et al.* 2005). Felsic to intermediate intrusions followed at *ca* 2520 to 2500 Ma, and the sequence was subsequently deformed and metamorphosed in the early Paleoproterozoic Sleafordian orogeny (2480–2420 Ma).

The Proterozoic development is divided into two intervals from 2000 to 1690 Ma and from 1690 to 1500 Ma by Hand *et al.* (2007). The older interval is dominated by rift-basin development, sedimentation and emplacement of the Donington granitic batholith at *ca* 1850 Ma in the eastern Gawler Craton. This interval was terminated by the 1730 to 1690 Ma Kimban orogeny. The younger interval was dominated by igneous and volcanic activity. Fanning *et al.* (1988) provide U–Pb dates for deformed volcanics of 1790 and 1740 Ma, and show that the voluminous Gawler Range Volcanics (GRV) and correlatives were erupted over a short interval at 1592 ± 2 Ma, and later intruded by anorogenic granites of the Hiltaba Suite Granitoids (HSG) at $\sim$1575 Ma.

Outcrop is most abundant in the central Gawler Craton particularly on the western and south-western margins of the GRV. The HSG and GRV are thought to be derived from partial digestion of the crust by mantle plumes (Neumann *et al.* 2000). The GRV and HSG are regarded as comagmatic having compositions similar to A-type magmas (Giles 1988; Creaser & White 1991; Creaser 1996). Recently Budd (2006) and Skirrow *et al.* (2006, 2007) have demonstrated that the Hiltaba Suite and Gawler Range Volcanics include both I- and A-type compositions. In the eastern and central Gawler Craton there is a predominance of A-type granites and volcanics, whereas I- types are largely restricted to the central and western parts of the craton.

Although the GRV and HSG have been generally described as 'anorogenic' (Giles 1988; Drexel *et al.* 1993), this notion has been more recently challenged by the increasing recognition of syntectonic features of the HSG (Creaser 1996; Ferris *et al.* 2002; McLean & Betts 2003; Lyons *et al.* 2005; Hand *et al.* 2007). Nevertheless, the granitoids are typically circular in outcrop.

The GRV and HGS are closely related to widespread mineralisation within the Gawler Craton (Johnson & Cross 1995), such as the world-class Olympic Dam Cu–U–Au–Ag ore deposit (Haynes *et al.* 1995) and Pb–Zn–Ag mineralisation at Menninnie Dam (Roache 1996). Wyborn *et al.* (1998) distinguishes between mineral associations corresponding to the Roxby Downs-type and the Kokatha-type. The Roxby Downs-type granitoids are strongly oxidised, fractionated and are characteristically pink due to hematite dusting of the feldspar crystals. This type is associated with Fe-oxide–Cu–Au deposits (Budd *et al.* 1998). The Kokatha-type granitoids are more reduced than the Roxby Downs-type and may contain pyrite. This type is associated with vein-hosted Au ($\pm$Sn$\pm$Ag) and displays a subdued magnetic expression.

The GRV form a large dominantly felsic, bimodal volcanic province that can be subdivided into an upper and a lower group. The lower group corresponds to the more oxidised (Roxby Downs-type) and is of varied lithologies, gently to steeply tilted and contains dacite-rhyodacite-rhyolite lava domes and flows and some ignimbrites with thick, intercalated sequences of basaltic lavas. These rocks typically carry magnetite. The upper group comprises thick, subhorizontal, porphyritic dacite sheets, predominantely ignimbritic in origin, and is ilmenite and titanomagnetite-bearing (Stewart 1993; Allen *et al.* 2008) belonging to Budd *et al.* (1998)'s Kokatha-type.

Of the rocks sampled for this study the Charleston Granite, Cunyarie Rocks Granite, Bittali Rhyolite, Eucarro Dacite, Yannabie Rhyodacite and Paney Rhyolite represent the Roxby Downs-type, while the Kokatha-type is represented by the Buckleboo Granite, the Yardea Dacite, the Black Yardea Dacite, rhyolites and porphyritic rhyolites.

PREVIOUS PALEOMAGNETIC STUDIES

Chamalaun & Dempsey (1978) reported paleomagnetic results for the Gawler Range Volcanics. Alternating field (AF) demagnetisation isolated magnetisation moderately upward-directed to the northeast. Thermal demagnetisation was performed on a few selected specimens representing different lithologies which showed a single magnetic carrier with a Curie temperature below 600°C, consistent with that carrier being magnetite. The thermal results gave directions similar to those from the AF demagnetisation but no mean directions were reported. Chamalaun & Dempsey (1978) reported that no consistent magnetisation directions were observed after heating to 600°C. They carried out XRD studies on representative lithologies and reported that only diffraction lines from magnetite were observed. The XRD data are thus consistent with the thermal demagnetisation results in that there is no evidence of hematite making a contribution to the remanence which Chamalaun & Dempsey (1978 p. 259) found 'a little surprising as the acid volcanics are often red ... and hence could contain a significant amount of hematite.' Any hematite present is apparently too finely divided to show up on XRD or to retain a stable remanent magnetisation.

From AF demagnetisation, the mean direction that Chamalaun & Dempsey (1978) determined from 52 magnetically stable samples (from 62 samples collected at 22 sites) was declination $D = 34.3°$, inclination $I = -48.1°$, $\kappa = 15.1$, $\alpha_{95} = 5.2°$ and corresponding pole position at latitude $\lambda_p = 60.4°$S, longitude $\varphi_p = 50.0°$E, $dp = 4.4°$, $dm = 6.8°$. They also calculated a mean direction based on five lithological groups. Interestingly, the mean direction from each group is not significantly different from the means of other groups (this detail is returned to below). No directions of opposite polarity were found and no field tests were performed to constrain the age of the magnetisation.

Schmidt & Clark (1992, 2000) revisited the Gawler Range Volcanics and after stepwise thermal demagnetisation of 26 samples found a declination $= 32.3°$, inclination $= -43.2°$, $\kappa = 20.0$, $\alpha_{95} = 6.5°$. The results herein include these 1992/2000 results as a sub-set and add results from a further 43 GRV samples for a total of 69 samples.

NEW ROCK MAGNETIC AND PALEOMAGNETIC DATA

The Gawler Peninsula field excursion handbook published by the Geological Society of Australia's Structural Geology and Tectonics Specialist Group (Parker *et al.* 1988) proved to be a very useful guide to the geology of the Gawler Range Volcacnics and the Hiltaba Suite Granitoids. We also used a Magellan GPS navigation instrument and an Elliot susceptibility meter for guidance in sampling. A schematic geological map for the Gawler Craton with sampling localities is shown in Figure 1. Sample localities are also listed in Table 1.

A total of 69 oriented cores/blocks from 14 units of the Hiltaba Suite Granitoids and Gawler Range Volcanics were collected using a magnetic and sun compass apparatus. A portable rock drill was used to take 2.5 cm diameter cores except where rocks were extremely hard, in which case block samples were collected. Between three and nine specimens, 2.2 cm high × 2.5 cm in diameter were machined from the block samples and at least two from each core sample. Two specimens per block and one per drill core were subjected to detailed paleomagnetic and rock magnetic experiments.

Laboratory techniques

Routine paleomagnetic laboratory methods (Collinson 1983; Butler 1992) were employed. Remanent magnetisation was measured using a CTF three-axis cryogenic magnetometer at CSIRO North Ryde, NSW. Hysteresis parameters and thermomagnetic curves were measured using a Munich Variable-Field Translation Balance (VFTB: URL: <http://www.vftb.com/gallery.shtml>). The demagnetisers used include a Schoenstedt AF demagnetiser and an in-house carousel furnace housed in a 10-coil Helmholtz set that provides feedback controlled 'field-free' space to about ± 5 nT. All specimens were subjected to stepwise thermal demagnetisation of between 10 and 15 steps. An in-house susceptibility bridge was used to measure low field magnetic susceptibility and k–T curves (Ridley & Brown 1980). Anisotropy of magnetic susceptibility was measured using a DIGICO anisotropy spinner interfaced to a PC.

Rock Magnetism

Hysteresis properties of rocks are useful in inferring the average domain structure of magnetic particles in rocks. Such domain structures are important when

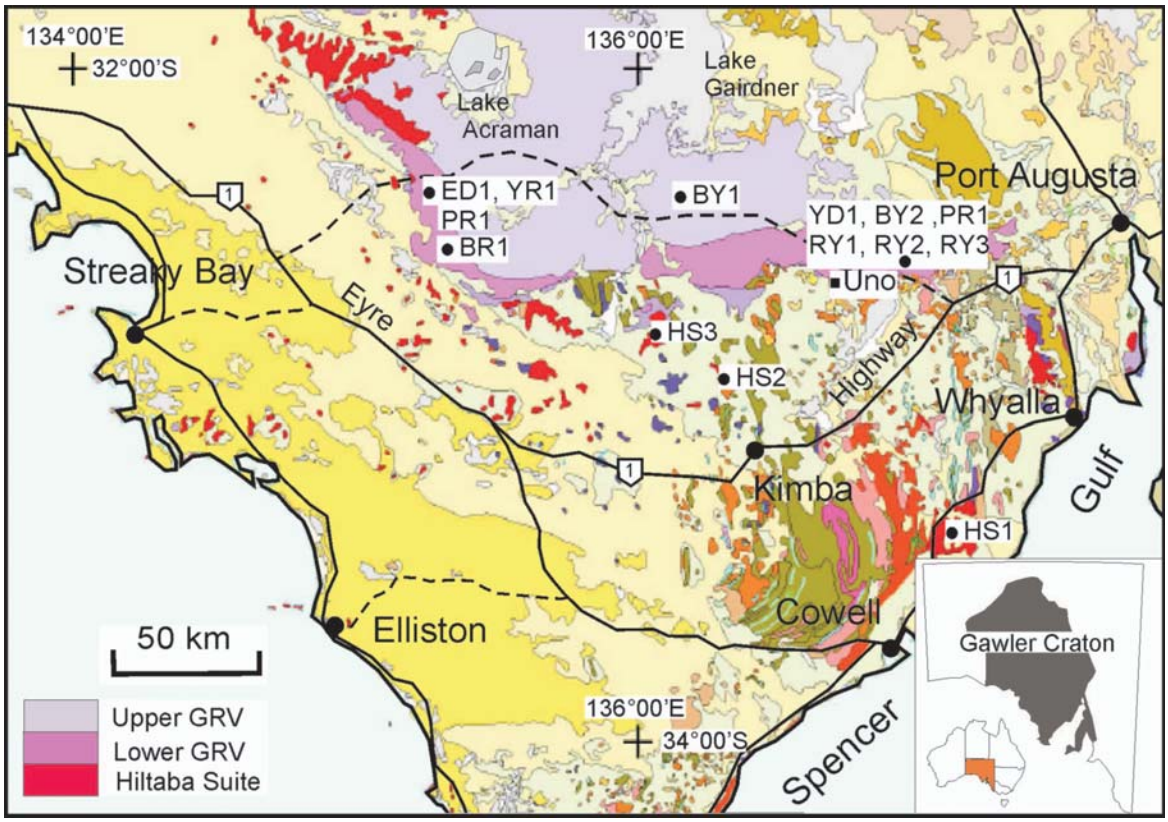


Figure 1 Map showing paleomagnetic sampling site in Hiltaba Suite Granitoids and Gawler Range Volcanics. Base map is from PIRSA. Site details are given in Table 1.

Table 1 Mean NRM directions and intensities and susceptibilities.

Site	N	Rock Unit (location)	Declination[a]	Inclination	Int (mAm^{-1})	k ($\times 10^{-5}$ SI)
HS1	15	Charleston Granite (33°22.81′S, 137°03.33′E)	11.2	−65.7	249	3357
HS2	18	Cunyarie Rocks (32°55.95′S, 136°17.75′E)	306.3	−57.6	2402	740
HS3	26	Buckleboo Granite (32°49.56′S, 136°04.32′E)	94.9	−11.8	357	154
BR1	12	Bittali Rhyolite (32°39.30′S, 135°21.84′E)	346.1	35.7	37.2	15.4
ED1	11	Eucarro Dacite (32°39.36′S, 135°40.02′E)	337.5	−9.3	2841	335
YR1	14	Yannabie Rhyodacite (32°37.86′S, 135°37.56′E)	313.4	−70.6	498	578
PR1	9	Paney Rhyolite (32°37.14′S, 135°44.76′E)	138.8	11.5	737	423
BY1	11	Black Yardea Dacite (32°31.38′S, 136°07.50′E)	283.7	−26.9	316	641
YD1	14	Yardea Dacite (32°36.10′S, 136°56.35′E)	276.1	−56.8	1759	355
BY2	15	Black Yardea Dacite (32°36.32′S, 136°56.28′E)	54.0	−43.9	3274	367
RY1	7	Rhyolite (32°36.74′S, 136°56.45′E)	12.4	−45.6	21.9	110
RY2	8	Rhyolite (32°36.74′S, 136°56.45′E)	20.1	−69.3	40.6	147
PH1	6	Porphyritic Rhyolite (32°36.94′S, 136°56.51′E)	0.7	−55.5	135	96.8
RY3	6	Rhyolite (32°36.94′S, 136°56.51′E)	3.7	−67.9	321	549

[a]Mean NRM is vector mean direction. N, number of specimens measured.
Sites with the acronym HS are Hiltaba Suite Granitoids, while all others are Gawler Range Volcanics. The acronyms in bold font are Kokatha-Type rocks while all others are of the Roxby Downs-Type.

considering the stability of remanent magnetisation and the likelihood that the magnetic remanence of a rock is original, or at least could withstand geological events such as burial subsequent to its formation. Day *et al*. (1977) used two ratios to infer domain structure. These are the ratios of the coercivity of remanence to the bulk coercivity (H_{cr}/H_c) and the saturation remanent magnetisation to the saturation magnetisation (M_{rs}/M_s). The bulk coercivity, H_c, saturation remanent magnetisation, M_{rs}, and saturation magnetisation, M_s, are derived from a hysteresis loop where the magnetisation is measured in a high field of up to 1 T. The paramagnetic response, which is linear to very high fields, and has been fitted and removed. H_{cr}, being a remanence related parameter, is measured in zero field after applying incrementally higher magnetic fields opposed to a (previously) saturated remanent magnetisation (see Figure 2a, b). The squarer the hysteresis loop, the more magnetically hard is a material. In terms of the above ratios this translates into high values (>0.05) of (M_{rs}/M_s) and low values (<4) of H_{cr}/H_c. Such values indicate grains with only a few domains, known as the pseudo-single domains (PSD), which, like single domains (SD), cannot be completely demagnetised. Acicular SD magnetite theoretically can possess M_{rs}/M_s values of 0.5. Alternatively, low values (<0.05) of (M_{rs}/M_s) and high values (>4) of H_{cr}/H_c suggest a high multiplicity of domains whose magnetisations to a large extent mutually cancel each other. This state is known as multidomain (MD), implying the existence of many walls within grains. Such walls can easily be moved through stress changes and thermal fluctuations at elevated temperatures. Rocks carrying MD magnetite grains are easily remagnetised by natural events.

A Day plot determined for samples of HSG and GRV is shown in Figure 2c. The majority of samples fall above the $H_{cr}/H_c = 4$ cutoff, identifying them as MD. The HSG samples in particular fall well into the MD region with one sample (Charleston Granite) having an extreme H_{cr}/H_c ratio of almost 20 indicating its remanence is very soft.

One sample (from site BYD1 in the Black Yardea Dacite) just falls in the PSD region, suggesting its remanence is reasonably stable. Nevertheless the characteristic direction of this unit is indistinguishable from those of MD units. While other GRV lithologies, especially the Kokatha-types, show M_{rs}/M_s values somewhat higher than the usual MD-PSD cutoff ratios of 0.05, remagnetised limestones from North America and Europe often exhibit elevated H_{cr}/H_c values, probably attributable to minor amounts of ultrafine superparamagnetic (SPM) grains (Dunlop & Özdemir 1997, p.323) and we assume that the presence of SPM or ultra-soft grains may be the case with the GRV. The boundaries separating SD to PSD, and PSD to MD are imprecise. Susceptibility versus temperature (k–T) curves (Figure 3a, b), and saturation magnetisation versus temperature (M_s–T) curves (Figure 3c, d), show that the only significant magnetic mineral present in both Roxby Downs and Kokatha types is almost pure end-member magnetite. The inflection near 350°C (Figure 3d) for the Black Yardea Dacite, is the only hint of a titaniferous phase. However, it is very minor. The irreversibility of the thermomagnetic curve, which indicates that this extra magnetic phase breaks down above ~400°C, is also consistent with the presence of some maghemite in the sample, which is commonly associated with incipient weathering of magnetite-bearing rocks.

The HSG and GRV display a range of properties from moderately to weakly magnetic (Table 1). The most intensely magnetic unit was the Kokatha-type Black Yardea Dacite (BYD2) with a natural remanent magnetisation (NRM) of 3.2 Am^{-1} and susceptibility of about 0.0036 SI. The least intensely magnetic unit was the Roxby Downs-type Bittali Rhyolite (BR1) with an NRM of only 37 mAm^{-1} and a susceptibility of only 0.00015 SI.

Roxby Downs-type Hiltaba Suite Granitoids, the Charleston Granite (HS1) and Cunyarie Rocks (HS2) are both quite magnetic, although the former possesses a moderate susceptibility but only a weak remanence, while the latter has a moderate remanence of 2.4 Am^{-1} but a low susceptibility of about 0.0075 SI. No consistent

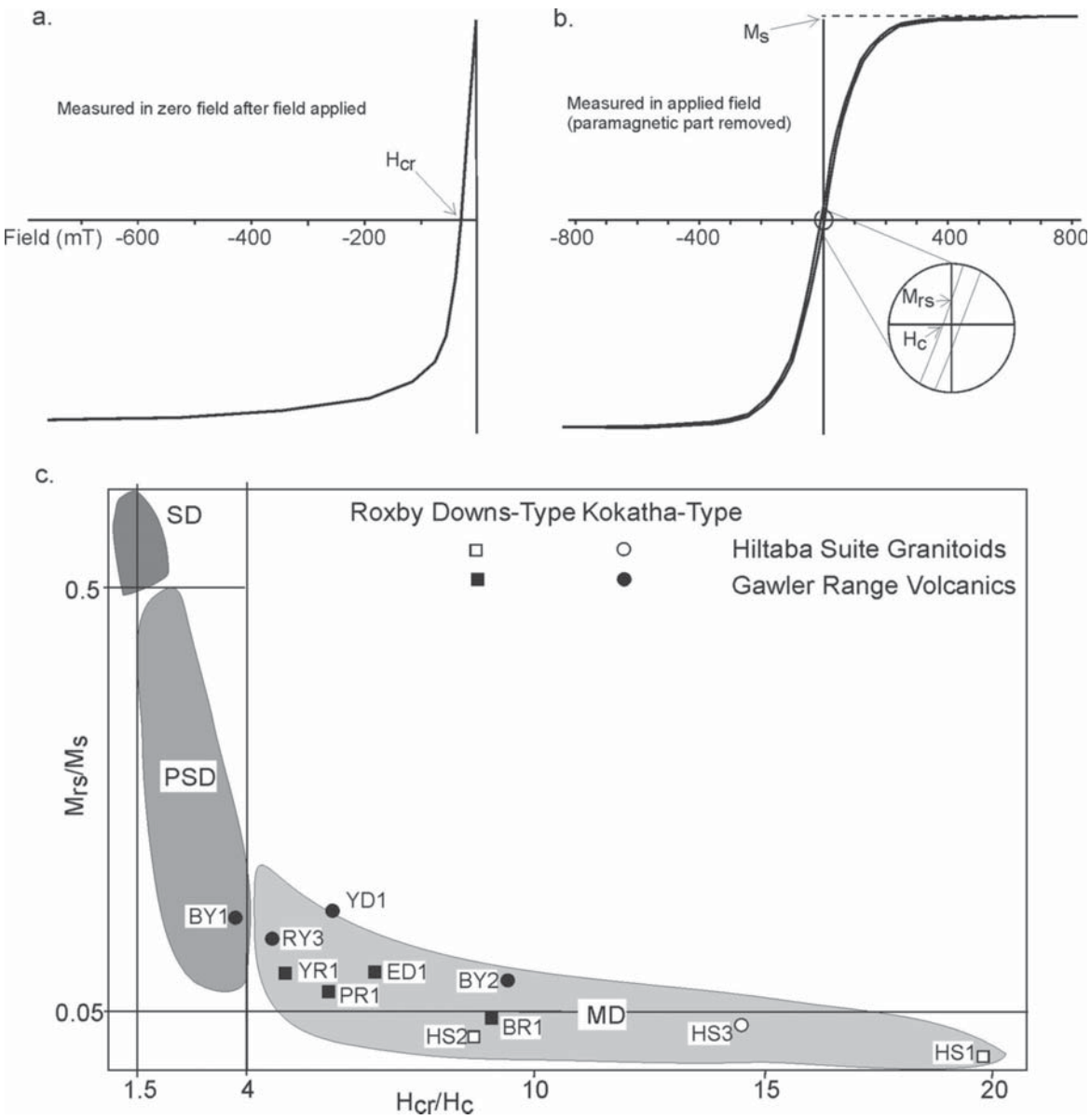


Figure 2 Rock magnetic data, (a) coercivity of remanence, H_{cr}, determined from back-field experiment, (b) hysteresis parameters determined from B-H experiment (after paramagnetic correction), and (c) Day-plot correlating M_{rs}/M_s and H_{cr}/H_c to identify magnetic domain structure.

direction of NRM was observed in the Cunyarie Rocks unit despite the moderately strong remanence. The Kokatha-type Buckleboo Granite (HS3) is weakly magnetic with a remanence of only 357 mAm^{-1} and a susceptibility of about 0.0015 SI. Thus, while the Roxby Downs-type HSG are magnetically stronger than the Kokatha-type Buckleboo Granite, the Kokatha-type GRV are magnetically stronger than the Roxby Downs-type GRV. Remembering that it was the Kokatha-type GRV that show the higher M_{rs}/M_s ratios suggests that of all the rocks sampled these would seem the most likely to retain any vestige of original thermoremanent magnetisation (TRM).

Overall magnetic properties are not diagnostic in distinguishing between Roxby Downs or Kokatha-type rocks. The MD nature of these carriers and, except for a minor titanomagnetite component in the Kokatha-type Black Yardea Dacite, their apparently uniform composition of pure end-member magnetite prevent the use of hysteresis or thermal techniques from differentiating their magmatic parentages.

Anisotropy of magnetic susceptibility (AMS) has also been investigated and results from all sites are listed in Table 2 (HSG) and Table 3 (GRV). The AMS is generally about 2% in the volcanics and about 4% in the granitoids. It is known that the DIGICO AMS delineator

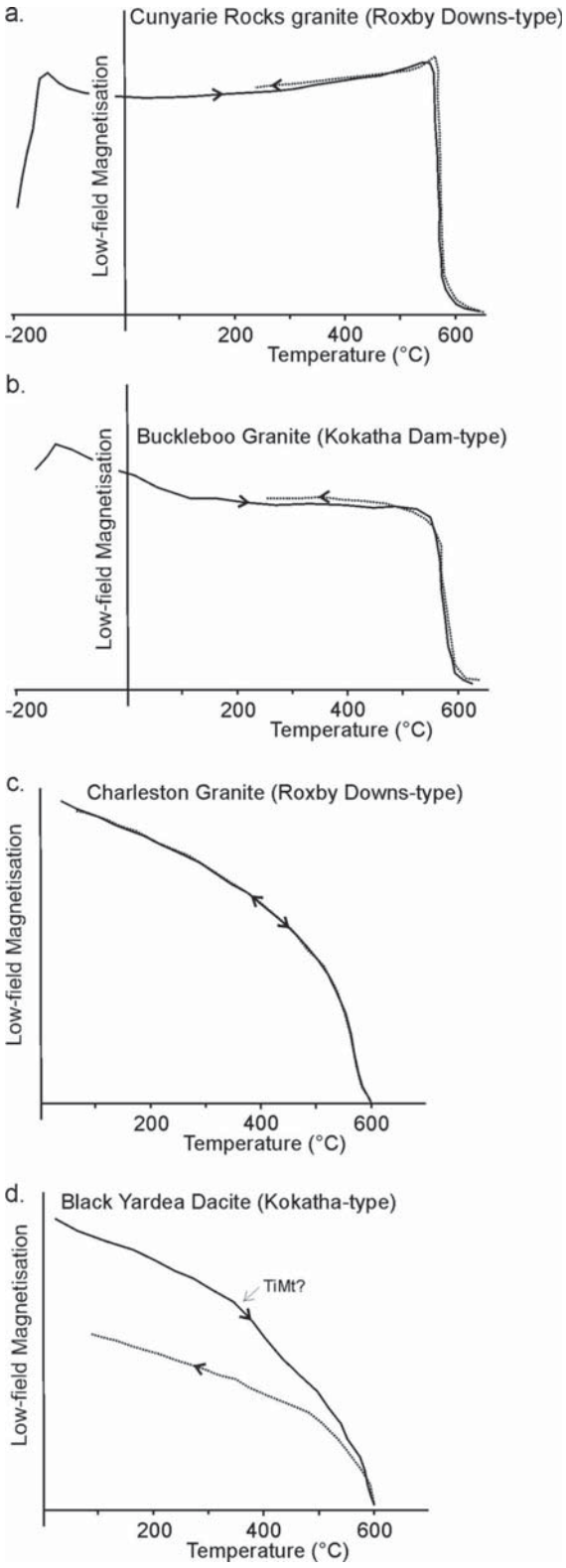


Figure 3 Susceptibility versus temperature (k–T) curves (a, b), in low field (~1 mT), and (c, d) saturation magnetisation (M_S–T) curves in fields up to 0.8 T for (a, c) Roxby Downs types and (b, d) Kokatha types.

can produce a lineation bias towards (cylindrical) specimen axes (Heller & Schultz-Krutisch 1988; Aubourg *et al.* 2004) but this is for weak fabrics and at the time this was the only instrument at our disposal. There is a mixture of prolate and oblate ellipsoids in both sets. The t factor in Tables 2 and 3 is positive for oblate ellipsoid and negative for a prolate ellipsoid. There do not seem to be any well-developed axes that are consistent between sites. The AMS of the GRV is low and a much larger study would be required to discern any patterns present (see McPhie *et al.* 2008 below). While the AMS of the HSG is somewhat higher than that of the GRV, there do not seem to be any consistent axes. A much higher number of sites may disclose some regional trends but there do not appear to be any trends present for the three granitoid sites in this study.

McPhie *et al.* (2008) studied the AMS of the Eucarro Rhyolite to locate source vents. They found foliations strike NW and dip steeply SW or NE respectively, depending whether near the base or top of the flow. They successfully identified two 'lobes' of the rhyolite, one west-central that flowed from the SW and the other to the east that flowed from the N. The only Eucarro Rhyolite site collected here (ED1: Table 3) has a 3% foliation dipping steeply to the north. This single result compares well with McPhie *et al.* (2008)'s site EF15 (their table 1 and figure 5a). Both their site EF15 and our site ED1 were from the west-central lobe.

Characteristic magnetisations

Samples were subjected to thermal demagnetisation, sometimes with alternating field (AF) pre-treatment for high intensities (see Figure 4a). This treatment was successful in removing very intense components that may have been the result of lightning. After applying principal-component analysis (PCA) to demagnetisation data from the GRV, a consistent high temperature remanence directed up to the north to north east was found (Table 4). This component showed a spread of unblocking temperatures from as low as 200°C upward, and was not fully demagnetised until 580°C which is consistent with the Curie point of magnetite. This is typical of pure end-member magnetite and especially typical of MD magnetite, which can have a tail of unblocking temperatures up to the Curie point even for partial TRMs acquired at much lower temperatures (Dunlop & Özdemir 1997 p. 248). Directions from HSG samples after PCA analysis resulted in scattered directions around the present or dipole field and no geological significance is attributed to them.

Orthogonal projections of the demagnetisation vectors for representative GRV samples are shown in Figure 4. Sample YD01c1 (Figure 4a) exhibits a large soft component that is removed by two AF steps of 30 mT and 60 mT before complete thermal demagnetisation is achieved after 580°C. In this sample the soft AF component has the same direction as the high temperature component, suggesting that the high intensity is not the result of lightning but is genuinely chacteristic of this unit. Other samples (Figure 4b–f)

Table 2 Summary of anisotropy of magnetic susceptibility for Hiltaba Suite Granitoids.

Site N	Declination (°)	Inclination (°)	k ($\times 10^{-5}$ SI)	A	L	F	t
HS1 15							
k_1	326	66	3416	1.044	1.009	1.034	0.564
k_2	201	14	3384				
k_3	106	18	3271				
HS2 18							
k_1	18	21	755.2	1.042	1.021	1.020	−0.025
k_2	278	23	739.7				
k_3	145	56	725.1				
HS3 25							
k_1	169	16	157.5	1.037	1.024	1.013	−0.280
k_2	75	12	153.9				
k_3	311	68	151.9				

N, number of specimens; k_1, k_2, k_3, maximum, intermediate, minimum susceptibility; A, anisotropy (k_1/k_3); L, lineation (k_1/k_2); F, foliation (k_2/k_3); t, shape parameter ([(2ln (k_2/k_3)/ln (k_1/k_3)] − 1.

Table 3 Summary of anisotropy of magnetic susceptibility for Gawler Range Volcanics.

Site N	Declination (°)	Inclination (°)	k ($\times 10^{-5}$ SI)	A	L	F	t
BR1 12							
k_1	107	13	15.55	1.011	1.005	1.005	−0.060
k_2	11	23	15.45				
k_3	225	61	15.37				
ED111							
k_1	299	59	340.5	1.035	1.005	1.031	0.718
k_2	90	26	338.9				
k_3	186	12	328.8				
YR1 14							
k_1	116	19	583.8	1.019	1.012	1.007	−0.241
k_2	270	68	576.9				
k_3	23	8	572.6				
PR1 9							
k_1	207	66	427.7	1.018	1.013	1.005	−0.412
k_2	78	15	422.3				
k_3	343	17	420.1				
BY1 11							
k_1	330	52	645.2	1.011	1.010	1.002	−0.658
k_2	80	14	639.5				
k_3	181	33	638.3				
YD114							
k_1	269	63	360.8	1.029	1.015	1.014	−0.034
k_2	104	25	355.6				
k_3	11	6	350.7				
BY2 15							
k_1	289	53	369.6	1.011	1.009	1.002	−0.670
k_2	159	25	366.4				
k_3	57	24	365.7				
RY1 7							
k_1	301	54	110.7	1.014	1.012	1.002	−0.718
k_2	124	34	109.4				
k_3	33	1	109.2				
RY2 8							
k_1	277	1	149.1	1.036	1.019	1.017	−0.030
k_2	7	20	146.3				
k_3	183	69	143.8				
PH1 6							
k_1	332	54	97.5	1.022	1.010	1.012	0.132
k_2	11	23	96.7				
k_3	225	61	96.2				
RY3 6							
k_1	338	29	527.0	1.024	1.003	1.020	0.668
k_2	68	8	524.9				
k_3	173	58	514.5				

N, number of specimens; k_1, k_2, k_3, maximum, intermediate, minimum susceptibility; A, anisotropy (k_1/k_3); L, lineation (k_1/k_2); F, foliation (k_2/k_3); t, shape parameter ([(2ln(k_2/k_3)/ln(k_1/k_3)] − 1.

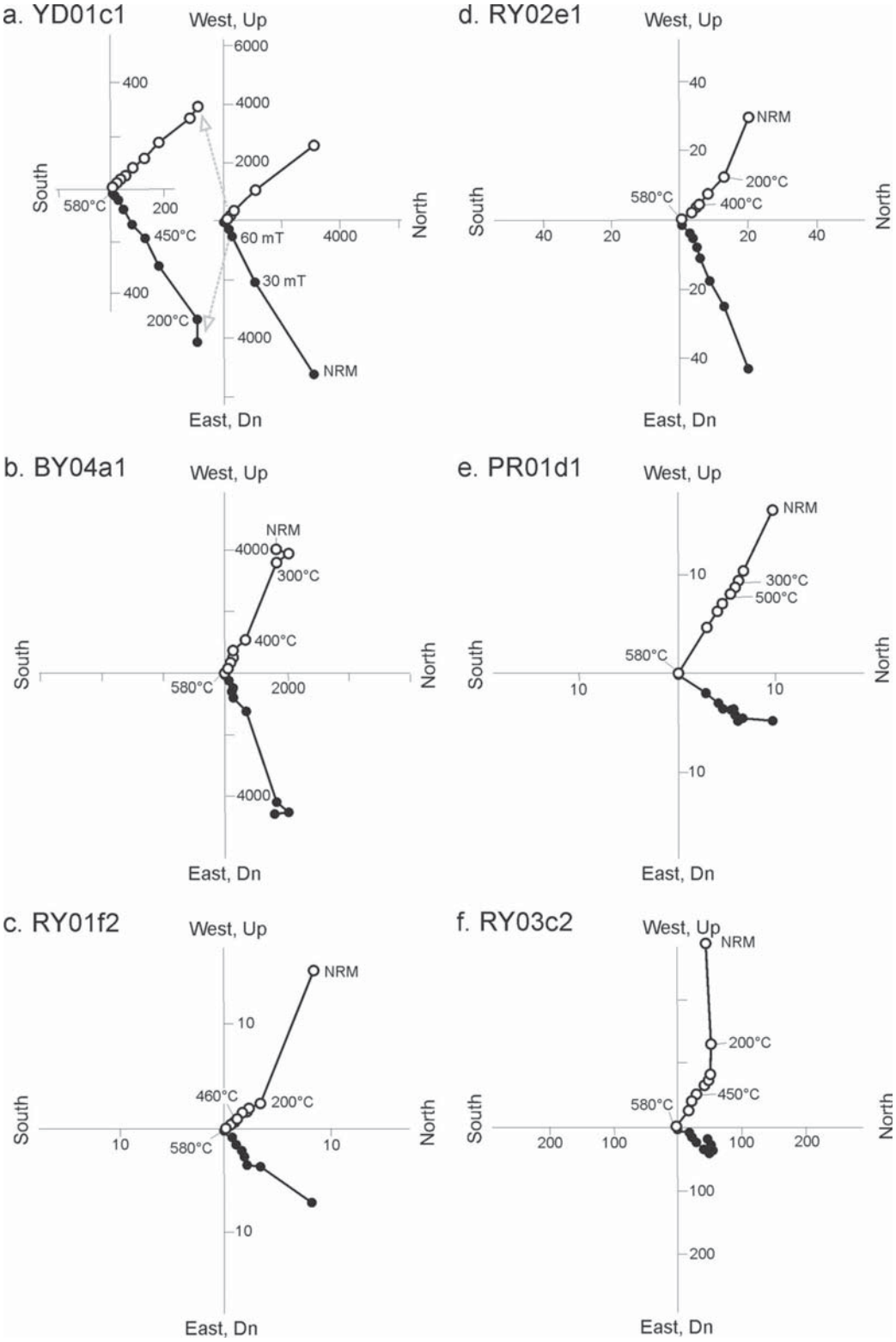


Figure 4 Orthogonal projections in geographic coordinates, not corrected for tectonic tilt, of remanence vectors after stepwise alternating field (AF in mT) and thermal demagnetisation (°C). Solid circles plot on the horizontal plane and open circles on the vertical plane. Axes are mAm^{-1}, temperature in °C. See text for details.

Table 4 Paleomagnetic summary for Gawler Range Volcanics (~32.0° S, ~136.0° E).

Site	N	D_h(°)	I_h(°)	DDA(°)	Dip(°)	D_b(°)	I_b(°)	k	α_{95}(°)
BR1	4	24.8	−15.6	–	0	24.8	−15.6	18.2	22.2
ED1	4	350.6	−15.7	–	0	350.6	−15.7	2.86	67.0
YR1	8	64.8	−25.9	–	0	64.8	−25.9	12.7	16.2
PR1	8	6.4	−50.8	–	0	6.4	−50.8	37.9	9.1
BY1	2	19.2	−58.5	–	0	19.2	−58.5	135	21.6
YD1	9	20.4	−54.8	–	0	20.4	−54.8	7.33	20.4
BY2	10	4.6	−49.8	005	85	185.4	−50.2	6.46	20.6
RY1	7	27.1	−28.2	005	85	145.4	−58.7	18.0	14.6
RY2	6	6.6	−54.1	005	85	183.8	−40.9	8.63	24.2
PH1	6	23.1	−53.5	005	85	170.0	−44.2	68.7	8.1
RY3	5	33.3	−51.6	005	85	161.5	−42.4	114	7.2
Site mean	11	20.5	−43.6			72.5	−73.1		
		$k = 13.1$	α_{95} (°) = 13.1			$k = 2.3$	α_{95} (°) = 34.0		
Sample	69	22.6	−44.9			94.1	−73.4		
Mean		$k = 7.2$	α_{95} (°) = 6.9			$k = 2.3$	α_{95} (°) = 12.8		
Fold test			$\xi_2 = 3.12$				$\xi_2 = 57.1$		

In situ pole position: $\lambda_p = 69.5°$S, $\varphi_p = 36.1°$E, dp = 5.5°, dm = 8.7°.
DDA, down-dip azimuth of bedding; *N*, number of samples showing ChRM; α_{95}, half-angle of the 95% confidence cone for the mean direction; D_h and I_h, *in situ* declination and inclination; D_b and I_b, declination and inclination with respect to bedding; k, precision parameter of Fisher (1953) distribution for mean directions (not to be confused with k in Tables 1–3 which stands for magnetic susceptibility); dp, dm, semi-axes of error ellipse around the pole of probability of 95%. ξ_2 is the fold test statistic (McFadden 1990) for which the 0.05 significance point ($N = 69$) is 9.66.

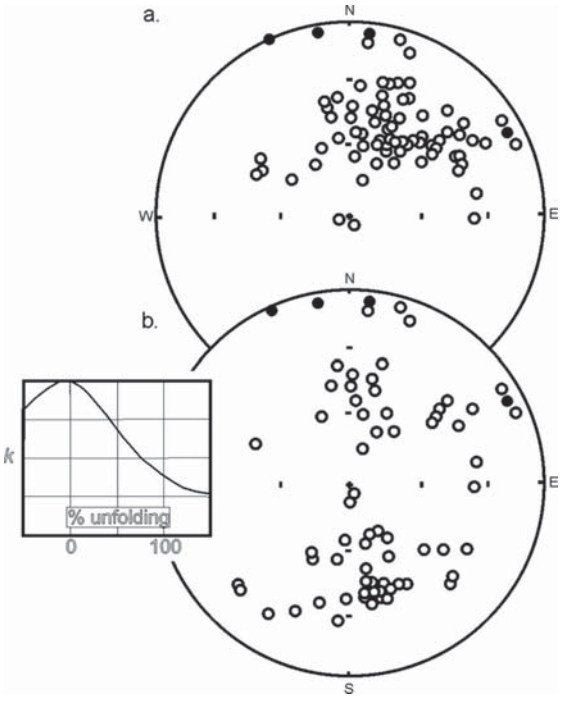


Figure 5 Equal-area projections of ChRM directions isolated by stepwise AF and thermal demagnetisation. Solid circles plot on the lower hemisphere and open circles on the upper hemisphere: (a) present-day coordinates, and (b) after structural correction. The tighter cluster of the former reflects a negative fold test at 95% confidence (McFadden 1990).

show very similar high temperature components directed to the NE with moderate negative (upward) inclinations (Table 4).

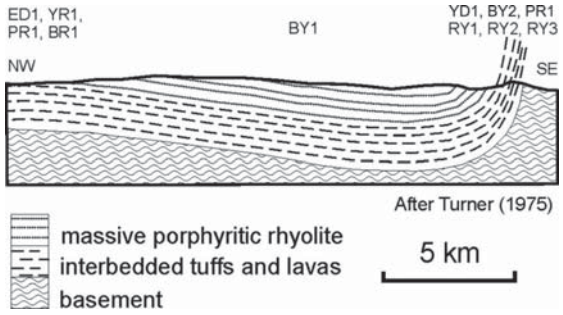


Figure 6 Geological cross-section from NW to SE showing steeply inclined strata near Uno (after Turner 1975). Sites YD1, BY2, PR1, RY1, RY2 and RY3 are located close to the cross-section just ENE of Uno (see Figure 1). Other sites are projected onto the cross-section to indicate their structural positions. No vertical exaggeration.

Directions of isolated components are presented as equal area projections (Figure 5). A paleomagnetic fold test (McFadden 1990) is possible for the GRV because the volcanics are dipping steeply to the north at Uno. The origin of the tilting is uncertain, although Turner (1975) emphasises that the attitude is considerably steeper than the angle of repose of 'fluidal lavas.' In fact, some parts are overturned and Turner (1975 p. 111) concludes that the 'major control... is tectonic and the structure... formed by peripheral upturning as a result of differential subsidence along the Uno Fault.' Figure 6 (after Turner 1975, figure 11) is a section from the NW to SE, with GRV sampling sites projected onto it. The volcanics, mainly rhyolites, at the SE end near Uno, are dipping at 85° to 005° TN. The statistic to test the correlation of individual sample directional deviation from the mean with the structural dip azimuth, ξ_2, is 3.12

compared with the 0.05 significance point of 9.66. This indicates rejection of a correlation of directions with their deviation from the mean and no reason to suspect that the magnetisation has not been acquired *in situ*. On the other hand, after applying the tilt correction to restore the strata to local horizontal, $\xi_2 = 57.1$ indicating a very strong correlation between sample direction and structure. The fold test is therefore strongly negative and we conclude that the GRV magnetisations have been acquired after the volcanics were tilted at Uno. Whether the structure formed soon after deposition of the conformable series or not, it is clear that any primary TRM has been overprinted.

Chamalaun & Dempsey (1978 p.260) noted that the mean directions for each of their five groups were very similar and interpreted this to indicate 'relatively little time difference between the units.' An alternative interpretation is that all the units have

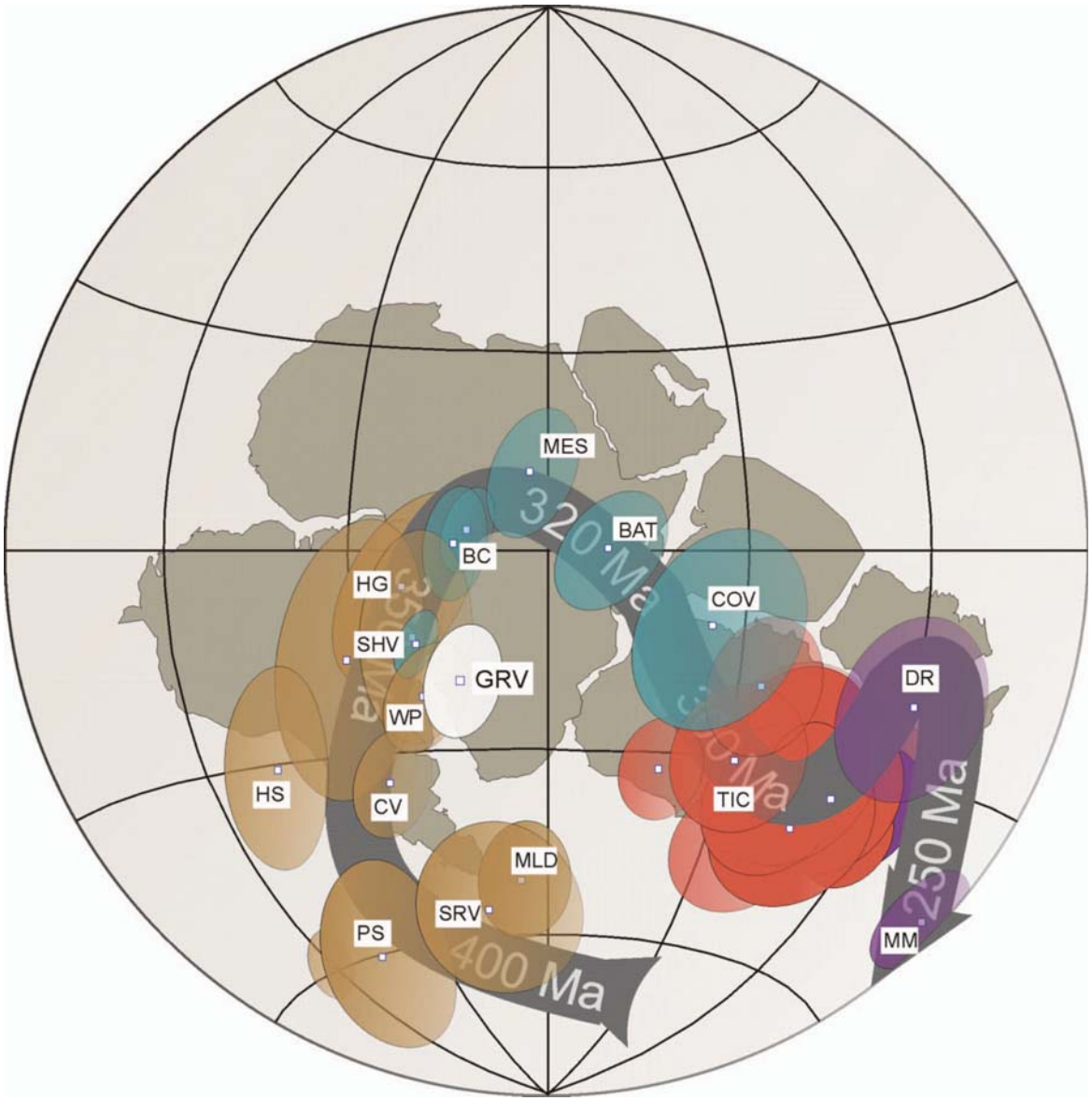


Figure 7 Late Paleozoic APWP for Australia (Table 5) in African coordinates using the Euler pole 22.7°N, 119.2°E and Euler angle of –54.8°. Colours of pole error ovals are coded in accord with the Commission for the Geological Map of the World (URL: <http://ccgm.free.fr/couleur_b_gb.html>). CMYK codes: Devonian, brown/tan (20/40/75/0); Carboniferous, turquoise/green (60/15/30/0); Permian, burnt sienna (5/75/75/0); Triassic, lilac/mauve (50/80/0/0). Mnemonics: SRV, Snowy River Volcanics; PS, Parke Siltstone; MLD, Mt Leyshon Dykes; HS, Hermannsburg Sandstone; CV, Comerong Volcanics; WP, Worange Point Fm; SHV, Silver Hills Volcanics; HG, Hervey Group; BC, Brewer Conglomerate; MES, Mt Eclipse Sandstone; BAT, Bathurst Batholith; COV– Connors Volcanics; TIC, Tuckers Igneous Complex; DR, Dundee Rhyodacite; MM, Milton Monzonite. Mnemonics for all poles plotted not shown. (after Clark & Lackie 2003).

Table 5 Mid-Paleozoic to Early Mesozoic pole positions for Australia.

Formation	Lomagage (Ma)	Himagage (Ma)	*N*	Declination (°)	Inclination (°)	α_{95} (°)	Lat_s (°)	$Long_s$ (°)	Lat_p (°)	$Long_p$ (°)	dp (°)	dm (°)	Refno	Resultno
Dundee Rhyodacite	242	252	39	281.5	82.8	6.5	−29.5	151.8	25.8	316.3	12.4	12.7	2566	6618
Dundee Ignimbrite	242	252	40	163.5	86.0	3.3	−29.3	151.9	36.9	334.8	6.5	6.5	2566	6619
Moonbi and Walcha Road	244	253	133	137.2	86.0	5.7	−31.1	151.3	36.8	338.0	11.2	11.3	3267	8413
Patonga Claystone	245	251	94	125.5	−87.5	3.9	−33.5	151.5	30.4	326.9	7.8	7.8	1610	1096
Milton Monzonite	245	260	39	59.9	78.3	3.6	−35.3	150.4	22.1	350.9	7.0	7.0	1238	239
Newcastle Range Volcanics	245	290	27	207.2	80.3	7.8	−18.3	143.77	30.9	319.7	16.8	16.8	3561	9057
Gerringong Volcanics	251	260	43	232.0	81.0	6.0	−34.6	150.8	44.0	312.0	11.0	11.0	995	1852
Gerringong Volcanics	260	271	19	67.0	81.0	11.0	−34.7	150.8	27.0	349.0	21.0	21.0	821	1823
Boggabri Volcanics	260	299	46	356.1	−78.0	5.5	−30.6	150.0	53.6	332.6	9.7	10.4	3639	9206
Mount Painter Combined	271	318	16	155.9	80.9	6.5	−30.2	139.2	46.0	329.5	12.0	12.5	2698	7019
Dotswood Redbeds	271	318	19	195.1	75.1	8.2	−19.8	146.4	46.1	315.6	13.7	15.0	406	1878
Mount Leyshon Complex	280	292	70	196.3	77.4	3.4	−20.3	146.3	43.2	317.3	6.0	6.4	3262	8401
Silver Hills Volcanics	280	320	16	220.2	67.7	7.0	−23.5	147.5	48.9	289.1	9.8	11.7	3266	8409
Star of Hope Fm Volcanics	280	320	25	143.4	80.7	4.7	−20.6	147.2	34.6	340.2	8.7	9.1	3266	8411
Featherbed Volcanics	280	305	166	200.6	75.0	5.1	−17.1	144.8	43.0	311.7	8.5	9.3	3266	8412
Tuckers Igneous Complex	283	291	78	188.7	75.2	3.8	−20.1	146.5	47.5	323.0	6.0	6.6	3531	8967
Bulgonunna Volcanics	293	305	27	204.8	78.6	9.6	−21.0	147.4	40.4	315.5	17.2	18.2	3330	8531
Brewer Conglomerate	300	340	4	184.4	74.3	6.2	−23.9	133.7	53.1	310.0	10.2	11.2	2726	7090
Connors Volcanics	310	320	36	228.0	63.0	10.0	−22.5	149.3	46.0	280.0	13.0	16.0	3265	8406
Mount Eclipse Sandstone	310	330	167	215.5	82.6	10.1	−22.3	131.3	33.8	301.2	19.2	19.7	2185	5709
Mount Eclipse Sandstone	310	330	65	226.7	82.3	6.2	−22.6	132.0	32.1	299.5	11.9	11.9	2866	7472
Newcastle Range Volcanics	317	325	78	191.4	61.6	5.1	−18.39	143.75	63.8	304.5	7.6	7.6	3561	9056
Star of Hope Fm Volcanics	318	359	6	352.4	2.6	18.4	−20.6	147.2	66.9	127.5	9.2	18.4	3266	8410
Bathurst Batholith	319	332	198	53.4	−50.2	7.6	−33.5	150.0	45.3	251.9	6.8	10.2	3264	8405
Retreat and The Overprint	320	340	34	208.1	47.8	4.5	−22.9	147.5	64.0	259.7	11.5	11.5	3622	9164
Silver Hills Volcanics	326	380	11	202.3	14.1	5.3	−23.5	147.5	63.1	203.8	2.8	5.4	3266	8408
Mount Eclipse Sandstone	330	345	69	237.3	37.7	8.6	−22.6	132.0	37.6	232.6	8.7	8.7	2866	7471
Hervey Group	359	375	56	29.2	−19.3	15.5	−33.2	148.5	54.4	204.1	8.4	16.2	1579	1031
Reef Complex I, Canning	359	385	89	42.7	−28.9	7.8	−18.3	125.6	49.1	218.0	4.7	8.6	1345	452
Brewer Conglomerate	359	375	33	224.9	29.1	6.0	−23.9	133.7	47.1	221.0	6.4	6.4	2726	7089
Worange Point Formation	359	375	63	21.1	−39.6	10.7	−36.9	149.9	67.9	208.6	10.9	10.9	2191	5722
Retreat Batholith	366	389	31	345.8	−6.7	3.8	−23.5	147.6	66.0	110.7	9.1	9.1	3622	9162
Reef Complex II, Canning	367	380	52	27.5	−19.9	15.2	−18.0	125.5	62.0	203.2	14.6	14.6	2942	7659
Hermannsburg Sandstone	375	392	24	21.0	−6.9	21.0	−24.0	133.0	61.0	180.9	15.6	15.6	2574	6649
Parke Siltstone	375	392	9	2.9	10.6	14.9	−23.6	132.1	60.9	138.1	7.6	15.1	2574	6650
Comerong Volcanics	380	390	52	0.4	−38.7	7.4	−35.5	150.0	76.9	150.7	7.2	7.2	1565	1003
Mount Leyshon Dev Dykes	382	415	25	349.7	−46.5	7.6	−20.0	146.5	78.0	18.8	6.3	9.8	3262	8404
Mereenie Sandstone	398	461	11	50.0	−24.5		−23.8	133.0	41.5	220.5	10.5	10.5	192	1904
Snowy River Volcanics	398	416	47	340.1	−59.2	9.7	−37.4	148.3	74.3	42.7	10.9	14.5	1365	486
Georgina Basin Carbonates	398	416	70	330.6	−3.9	5.8	−22.4	140.3	54.9	81.6	2.9	5.8	3616	9152
Chatswood Lst Ninmaroo Fm	398	416	8	333.5	−17.7	15.5	−22.6	140.3	61.2	74.1	8.3	10.1	3082	8084

(*continued*)

Table 5 (*Continued*).

Formation	Lomagage (Ma)	Himagage (Ma)	N	Declination (°)	Inclination (°)	α_{95} (°)	Lat_s (°)	$Long_s$ (°)	Lat_p (°)	$Long_p$ (°)	dp (°)	dm (°)	Refno	Resultno
Mereenie Sandstone	398	461	51	292.0	20.0	25.6	−24.0	132.0	15.7	62.7	23.7	23.7	2574	6648
Mount Leyshon Silurian Dykes	422	428	14	289.2	−21.6	8.3	−20.1	146.4	−21.7	231.9	4.6	8.7	3262	8418
Ravenswood Batholith	422	428	51	285.4	−18.0	7.1	−20.1	146.4	−17.5	232.7	3.8	7.3	3262	8403
Georgina Basin Carbonates	428	461	115	272.4	−27.6	5.3	−22.4	140.3	−8.0	216.8	2.6	5.1	3616	9153
Theresa Creek Volcanics	428	488	37	62.4	59.5	4.6	−22.8	147.6	4.1	188.9	14.0	14.0	3622	9163
Chatswood Lst Ninmaroo Fm	428	444	17	199.9	−7.5	11.4	−22.6	140.3	57.2	359.3	11.5	22.7	3082	8083
Chatswood Lst Ninmaroo Fm	479	495	64	274.5	4.3	7.4	−22.6	140.3	−3.1	234.1	7.4	14.8	3082	8082

Late Paleozoic paleomagnetic poles for Australia in African coordinates using the Euler pole 22.7°N, 119.2°E and Euler angle of −54.8° (Figure 7).
Lat_s (°), site latitude; $Long_s$ (°), site longitude; Lat_p (°), pole latitude; $Long_p$ (°), pole longitude.
Refno, reference number; Resultno, result number from the Global Palaeomagnetic Database (GPMDB; McElhinny & Lock 1993, 1996).
Lomagage, minimum magnetic age; Himagage, maximum magnetic age.

been remagnetised over a similar time interval. Given the negative fold test this possibility seems the more likely and we have opted to base our mean direction on unit samples. The *in situ* mean direction from 69 samples is declination = 22.6°, inclination = −44.9°, $\alpha_{95} = 6.9°$ and the corresponding pole position is latitude $\lambda_p = 69.5°$S, longitude $\varphi_p = 36.1°$E, dp = 5.5°, dm = 8.7°.

This result can be compared with that of Schmidt & Clark (1992, 2000) from 26 samples of declination = 32.3°, inclination = −43.2°, $\alpha_{95} = 6.5°$ using the F-test formulated by McFadden & Lowes (1981). For this comparison the degrees of freedom are 2 and 186 for which $F_{2,186} = 3.05$ for $p = 0.05$. The value of the comparative statistic is 1.17 so we may assume with 95% probability that there is no significant difference between these directions.

We may also compare this new result with that of Chamalaun & Dempsey (1978) for 52 samples of declination = 34.2°, inclination = −48.1°, $\alpha_{95} = 5.2°$ using the same F-test (McFadden & Lowes 1981). For this comparison the degrees of freedom are 2 and 238 for which $F_{2,238} = 3.03$ for $p = 0.05$. The value of the comparative statistic is 2.77, which again suggests that with 95% probability there is no significant difference between these directions. The new result is therefore not statistically different from either Chamalaun & Dempsey (1978)'s result based on AF demagnetisation or our earlier result based on fewer samples.

DISCUSSION

The paleomagnetic pole position from the GRV is plotted on Australia's Mid to Late Paleozoic apparent polar wander path (APWP) in Figure 7 following Clark & Lackie (2003). The poles that comprise the APWP are listed in Table 5. Since the GRV magnetisation is shown here to be an overprint, it is appropriate to estimate the age of the overprint from its position on the APWP using the nearest well-dated pole positions. As pointed out by Clark & Lackie (2003), the GRV poles calculated by Chamalaun & Dempsey (1978) and Schmidt & Clark (1992, 2000) lie close to the poles from the Hervey Group (375–359 Ma) and Silver Hills Volcanics (345–330 Ma). The GRV pole calculated here ($N = 69$) supersedes that of Schmidt & Clark (1992, 2000 $N = 26$) and falls a little to the south closer to the Worange Point Formation pole (375–359 Ma) in African coordinates. This difference is trivial and by either estimation (Schmidt & Clark 1992, 2000 or herein) the GRV magnetisations appear to be Late Devonian in age. Statistical tests described above indicate no significant difference between this new result and the previous results from the GRV. The 'key-pole' status afforded the GRV pole (Idnurm & Giddings 1988) must be abandoned.

It has long been recognised that many rock units throughout central and southern Australia have been remagnetised during the Late Paleozoic. This has often been attributed to the waning phase of the Alice Springs Orogeny (ASO), which seems to have lasted for a considerable time. McLaren *et al.* (2002) present thermochronological evidence from the Mount Painter area for denudation events probably associated with the ASO at 430, 400 and about 330 Ma. These events would also

have affected the Gawler Craton. Schmidt & Embleton (1987) have synthesised the ASO magnetic overprints, showing them to be variously pre-, syn- and post-folding in age. Many of the pole positions are apparently mid- to late Carboniferous in age, suggesting that they coincided with the waning phase of the ASO (<330 Ma).

On the basis of apatite fission-track ages (AFTA), Spikings *et al*. (2006) extend the effects of the ASO to the Mt Isa and Murphy Inliers of northeast Australia. There is increasing evidence that the ASO affected cratonic areas for thousands of kilometres from central Australia. Ferguson (1981) reported a mean of 331 ± 30 Ma for 13 AFTA for the Gawler Craton and AFTA for shattered Yardea Dacite from the Acraman impact site yield an age of 319 ± 19 Ma (data pers. comm. G. E. Williams 2000 courtesy of I. R. Duddy, Geotrack International, Melbourne). These early Carboniferous ages indicate when the Gawler craton cooled below the apatite fission track retention temperature (100 ± 20°C). The unimodal, negatively skewed distribution of track lengths reflects a slow, steady cooling history with the fission tracks marking the time ~100°C was reached, some tens of millions of years after attainment of the temperatures that are associated with the overprint magnetisations carried by the GRV. This is in agreement with Kohn *et al.* (2002) who concluded on the basis of AFTA that the denudation chronology for the Gawler Craton 'shows a fairly uniform rate of *ca* 10 m per million years over the last 300 million years' (p. 710). If linear, this implies that the Gawler Craton has undergone *ca* 3 km of erosion since 300 Ma; however, there is a reasonable likelihood that some periods saw higher denudation rates than others, especially the period immediately following uplift associated with the ASO.

The denudation model of Kohn *et al.* (2002) is based partly on the paleogeothermal gradient assumed. If this were elevated compared with those of the present day, then the calculated thickness of crustal erosion would be less. Nevertheless, assuming a higher paleogeothermal gradient for the Gawler Craton in the Late Paleozoic (Williams & Wallace 2003) suggests 2 km of erosion is not unreasonable.

CONCLUSIONS

The measured magnetic properties of the Hiltaba Suite Granitoids explain their highly variable airborne magnetic signatures. Cunyarie Rocks (HS2) displays a prominent anomaly, while the magnetic signature of the Buckleboo Granite (HS3) is flat. This is usually attributed to the Hiltaba Suite Granitoid subdivision of oxidised hematite/magnetite bearing Roxby Downs-type (Cunyarie Rocks granite) and ilmenite/titanomagnetite bearing Kokatha-type (Buckleboo Granite). However, rock magnetic tests here have identified magnetite as the overwhelming magnetic mineral, in both types of rocks, except for a minor phase of titanomagnetite in the Kokatha-type Black Yardea Dacite. As expected from their airborne magnetic signatures, the Buckelboo Granite is very much less magnetic than the Cunyarie Rocks granite, so while no rock magnetic evidence for titanomagnetite was found in the Buckelboo Granite, the amount is extremely small. Since ilmenite is not ferromagnetic, the tests applied here are not able to quantify the amount of ilmenite that may be present in the Kokatha-types.

No simple picture emerges from comparing Roxby Downs and Kokatha-type GRV, although the Yardea Dacite and Black Yardea Dacite (Kokatha-types) appear to be more magnetic than most Roxby Downs-types. However, one of these latter types, the Eucarro Dacite, is also quite magnetic.

The Gawler Range Volcanics at Uno are dipping almost vertically to the north but possess remanence directions similar to those found in almost flat lying volcanics to the west, and also to those reported by Chamalaun & Dempsey (1978). Clearly, the Gawler Range Volcanics have been magnetised after tilting. Comparison of the GRV pole position with Australia's mid to late Paleozoic apparent polar wander path suggests that the overprint magnetisations date from the late Devonian (375–360 Ma).

Apatite fission track ages indicate that (the present surface of) the Gawler Craton cooled through the ~100°C isotherm in the Late Carboniferous, some tens of millions of years after attainment of the peak temperatures and after the magnetic overprint of the Gawler Range Volcanics became blocked. The magnetic overprints and AFTA therefore each represent an isotherm records, albeit of different temperatures, of the cooling of the Gawler Craton after the widespread Alice Springs Orogeny.

ACKNOWLEDGEMENTS

We are grateful for discussions with John Parker and Terry Crabb in 1990 when fieldwork was first undertaken as part of AMIRA project 78/P96C. The authors are grateful to George Williams for providing unpublished material on AFTA from the Acraman meltrock and for many discussions. Dr Chris Klootwijk and Dr Robert Musgrave are thanked for their critical comments on the MS.

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Received 18 August 2009; accepted 3 December 2010

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